

SU Jiaji

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EDUCATION

JAN 2024	DOCTOR OF PHILOSOPHY IN STATISTICS Department of Statistics & Data Science <i>National University of Singapore</i> , Singapore THESIS ADVISOR: Dr. YAO Zhigang
JUN 2018	BACHELOR OF NATURAL SCIENCE IN STATISTICS School of Mathematical Science <i>Zhejiang University</i> , Hangzhou, China

PROFESSIONAL EXPERIENCE

MAR 2026 – CURRENT	HONGSHEN YOUNG FACULTY College of Mathematics and Statistics Chongqing University, Chongqing, China	
JUL 2025 – MAR 2026	INDEPENDENT RESEARCHER Chengde, Hebei, China	
OCT 2023 – JUL 2025	RESEARCH FELLOW Department of Statistics & Data Science National University of Singapore, Singapore	Advisor: Dr. YAO Zhigang
AUG 2022 – OCT 2023	STUDENT RESEARCH ASSISTANT Department of Statistics & Data Science National University of Singapore, Singapore	Advisor: Dr. YAO Zhigang

SELECTED ONGOING PROJECTS

SINCE JUN 2025	MANIFOLD FITTING FOR BIOLOGICAL SEQUENCE EMBEDDINGS - Extends our manifold fitting framework to high-dimensional embeddings produced by Transformer-based DNA/RNA language models. - Identifies low-dimensional structures that coherently organise sequences sharing biological functions, providing geometric insight into the “sequence language space.” - Incorporates the fitted manifolds into classification, clustering, and other downstream tasks, enhancing accuracy and interpretability over raw embeddings.
SINCE JAN 2024	LOW-DIMENSION STRUCTURE ANALYSIS BASED ON UK BIOBANK - Applies manifold-fitting methods to large-scale biomedical data, with a focus on the UK Biobank. - Aims to uncover low-dimensional structures within high-dimensional population health data. - Leverages these structures to improve population stratification and refine medical risk assessments.

PREPRINTS & PUBLICATIONS

ACCEPTED:

*Equal Contribution; #Correspondence

- *Li, B., *Su, J., *Lin, R., #Yau, S.-T., & #Yao, Z. (2025). Manifold fitting reveals metabolomic heterogeneity and disease associations in UK biobank populations. *Proceedings of the National Academy of Sciences*, 122(22), e2500001122.
- **Yao, Z., *Su, J., & **Yau, S.-T. (2024). Manifold fitting with cycleGAN. *Proceedings of the National Academy of Sciences*, 121(5), e2311436121.
- Su, J., #Yao, Z., Li, C., & Zhang, Y. (2023). A statistical approach to estimating adsorption-isotherm parameters in gradient-elution preparative liquid chromatography. *The Annals of Applied Statistics*, 17(4), 3476–3499.

PREPRINTS:

- Su, J. & #Yao, Z. (2025). Principal decomposition with nested submanifolds. *arXiv preprint arXiv:2502.10010*. (Biometrika RR).
- #Yao, Z., Su, J., Li, B., & Yau, S.-T. (2023). Manifold fitting. *arXiv preprint arXiv:2304.07680*. (JMLR under review).

TALKS & PRESENTATIONS

OCT 2025 Chongqing	TALK: ‘MANIFOLD FITTING AND PRINCIPAL NESTED SUBMANIFOLDS’ <i>College of Mathematics and Statistics, Chongqing University</i>
MAY 2025 Beijing	YMSC STATISTICAL SEMINAR <i>Yau Mathematical Sciences Center, Tsinghua University</i>
JAN 2025 Shanghai & Sanya	THE SECOND SYMPOSIUM OF GEOMETRY AND STATISTICS IN CHINA <i>Shanghai Institute for Mathematics and Interdisciplinary Sciences & The Tsinghua Sanya International Mathematics Forum</i>
OCT 2024 Singapore	INTERACTIONS OF STATISTICS AND GEOMETRY (ISAG) II <i>Institute for Mathematical Sciences, National University of Singapore</i>
JUN 2024 Shanghai	SUMMER SEMINAR SERIES: STATISTICS + GEOMETRY + X <i>Shanghai Institute for Mathematics and Interdisciplinary Sciences</i>
JUL 2023 Beijing	TALK: ‘ESTIMATING ADSORPTION-ISOTHERM PARAMETERS’ <i>School of Mathematics and Statistics, Beijing Institute of Technology</i>

VISITING

JUN 2024 – JUL 2024	VISITING RESEARCHER <i>Shanghai Institute for Mathematics and Interdisciplinary Sciences, China</i>
DEC 2024 – JAN 2025	VISITING RESEARCHER <i>Shanghai Institute for Mathematics and Interdisciplinary Sciences, China</i>

TEACHING

Teaching assistant at National University of Singapore:

Semester	Courses	
2021-22 Sem 2	ST2334	Probability and Statistics
	DSA1101	Introduction to Data Science
	ST2137	Statistical Computing and Programming
2021-22 Sem 1	ST5213	Advanced Categorical Data Analysis
	DSA1101	Introduction to Data Science
2020-21 Sem 2	ST2334	Probability and Statistics
	ST3232	Design and Analysis of Experiments
2020-21 Sem 1	ST2334	Probability and Statistics
	ST5225	Statistical Analysis of Networks
2019-20 Sem 2	ST2131	Probability
	ST3239	Survey Methodology
2019-20 Sem 1	ST4231	Computer Intensive Statistical Methods
2018-19 Sem 2	ST2131	Probability